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Inverse-designed compact and low-loss bending waveguides on anisotropic LNOI platform with mode-field analysis

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A B S T R A C T

Compact and low-loss bends are crucial for achieving high-density photonic integration on the Lithium niobate on insulator (LNOI) platform. Traditional design usually utilizes random search to minimize the transmission loss, which is time-consuming for large structures and limits further loss reduction by considering anisotropy on the LNOI platform. Here we propose a Fast Directional Bisection with Mode-Field Analysis for the design of bending waveguides on anisotropic LNOI platform. The bends are realized through adjusting the geometric parameters of the cascaded Generalized Euler segments with careful consideration of the anisotropic refractive index variation. Three bends with different equivalent radii are designed to demonstrate the versatility of the proposed method. A low-loss at 1550 nm (0.008 dB) and compact SWB (R_{eq} = 40.5 μm) has been experimentally demonstrated for the first time. Employing the same approach, we realize a compact and low-loss Optical delay line (ODL), with a total length of 3.64 cm and a footprint of 0.5 × 0.5 mm^2, achieving an average transmission loss of 0.43 dB/cm. It represents an 80 % reduction in comparison to the Euler-bend ODLs. The proposed design method and low-loss SWBs/ODLs held potential for advancing the future development of LNOI photonic integrated devices and circuits.

1. Introduction

Lithium niobate (LN) is renowned for its high refractive index, wide spectral transparency, efficient electro-optic effect, and significant χ(2) nonlinearity, making it an ideal material for a wide range of optical and electronic applications [1,2]. In recent years, with the Thin-film lithium niobite (TFLN) technology and the rapid advancements in micro-nano fabrication techniques, the Lithium niobate on insulator (LNOI) platform has become a key foundation for integrated photonic devices [3–5]. LNOI-based devices have demonstrated exceptional performance in electro-optic modulation [6–10], acoustic-optic modulation [11,12], as well as frequency conversion [13–15], offering distinct advantages such as high speed, high miniaturization and low power consumption. However, as the demand for optoelectronic systems with higher integration level and larger scale level grows, a critical challenge for LNOI platforms is how to minimize the transmission losses and footprints in various optical components [16–19]. As one typical component in photonic systems, bending waveguides are widely used to alter light propagation paths, reducing the physical space required for straight waveguides and enabling more compact optical circuit layouts, including optical delay lines (ODLs) [20–24], large-scale photonic switches [25–28], and on-chip photonic processors [29–31]. To realize a highly efficient large-scale optoelectronic system, a compact and ultra-low-loss bending waveguide is highly required on LNOI platform [16].

Traditional bending waveguide designs, often based on circular arcs, typically require large bending radii to maintain low-loss transmission. While this design strategy minimizes bending losses, it significantly increases the waveguide's footprint, limiting the overall integration density [20–24]. To address this limitation, recent innovations, such as subwavelength-assisted structures [32–34], air trenches [35,36], and inverse design Based on Directional Binary Search (DBS) [37], have sought to create more compact bends. However, these methods demand high-precision photolithography and tend to introduce extra scattering losses due to the complex subwavelength structures, which will accumulate in large-scale systems and ultimately decrease the overall performance. Alternative solutions, including continuous transition curves such as Euler bends [38] and Bezier bends [39], allow for more compact

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designs while maintaining low-loss transmission and better fabrication tolerance. Despite their advantages, the optimization of such curves is often limited by specific mathematical expressions such as Euler function and Bezier function. To overcome this limit, the concept of continuous free-form curve (FFC) has emerged, enabling the design of bending waveguides composed of multiple segments with independently adjustable curvatures [40–42]. The curves not only provide smaller dimensions but also achieve similar low-loss performance. However, these designs typically rely on random optimization techniques, which is time-consuming and computationally expensive [43]. To reduce the computation cost, a symmetric curve model has been applied to the FFCs design on the X-cut LNOI platforms [40], which however does not align with LN's inherent anisotropy. The anisotropic refractive index variations have a great influence on the bending loss, which brings about further challenges in achieving compact and low-loss design on LNOI platform.

In this work, we propose a Fast Directional Bisection with Mode-Field Analysis (FDB-MFA) method for the design of compact and low-loss single-mode bending waveguide (SWB) on anisotropic LNOI platform. Using the method, The SWB based on the FFC structure is achieved by adjusting the geometric parameters of cascaded Generalized Euler segments through mode-field analysis, while simultaneously accounting for the impact of refractive index variations on bending. We propose three FFC designs with different equivalent radii on the X-cut LNOI platform, featuring a slab thickness of 200 nm and a rib height of 200 nm. These designs outperform traditional Bezier, Euler, and Arc bending counterparts of same dimensions, demonstrating significant advantages in loss reduction. Experimental results show that the 90-degree single-mode bent waveguide with an equivalent radius of 40.5 μm has a remarkably low loss of only 0.008 dB at 1550 nm wavelength. The design allows for a manufacturing tolerance of up to ±200 nm. Additionally, we extend our investigation to a compact and low-loss FFC-bend ODLs on the same platform, featuring a footprint of approximately 0.5 × 0.5 mm^2 and a total length of 3.64 cm. The results show that the average transmission loss is as low as 0.43 dB/cm, which represents an 80 % reduction in comparison to the Euler-bend ODLs of the same size. It is expected that our study can play an important role in advancing the development of photonic integrated devices and circuits on anisotropic LNOI platform.

2. Design principle of FFC

Fig. 1(a) shows the perspective view of the proposed 90-degree bending waveguide based on FFC structure, designed using the FDB-MFA method. The waveguide structure is fabricated on a regular X-cut LNOI platform, with a LN core layer surrounded by a silica cladding. To better visualize the FFC structures, the upper oxide cladding is ignored. The FFC is divided into three key regions: two mode matching regions and one radiation suppression region. The mode matching region is designed to ensure efficient coupling between the straight and bending waveguides, in which the curvature radius is reduced from infinite to finite value by carefully shaping the curve in this region, the light smoothly transitions into the bending waveguide with minimal loss. Typically, a smaller curvature radius will cause more radiation leakage and thus results in higher losses. The radiation suppression region primarily addresses this issue by balancing the trade-off between minimizing the curvature radius and suppressing radiation losses, thereby ensuring efficient mode propagation within the small-radius region of the bending waveguide. The cross-section of the waveguide is illustrated in the inset of Fig. 1(a), where it is defined by a total film thickness T, an etch depth H, a bottom width W, and a sidewall angle α.

The schematic diagram of the FFC structure is illustrated in Fig. 1(b). The angle between the propagation direction of the light and the optical axis Y of the LN material is θ. The two edge lengths of the quadrilateral region occupied by the bending waveguide are labeled as R_y and R_z. Due to the anisotropic properties of the LNOI platform, the symmetry of the FFC is broken, so R_y and R_z are not always equal. Therefore, the equivalent radius R_{eq} is defined as the maximum of R_y and R_z, providing a consistent metric for comparison with conventional bending waveguide. The entire FFC is discretized into a series of cascaded segments, with the terminal curvature radius of each segment monotonically decreasing from the start of the bend to shrink the overall bending radius. Unlike previous circular arc structures, our proposed curve segment model is based on the Generalized Euler spiral, which will be explained in details latter. This new kind of curvature can provide greater design flexibility, allowing for the exploration of curvature distributions that cannot be realized with traditional arc-based designs, which is especially advantageous for the development of compact and low-loss bending free-form waveguides. As shown in Fig. 1(c), the i th segment is parameterized by its starting curvature radius RS_i, terminal curvature radius RT_i, the corresponding curvature center O_{is} and O_{it}, the turning angle $\Delta\theta_i$, along with a constant length ΔL and width W.

A detailed description of the construction of the Generalized Euler spiral is presented below. For modified Euler spiral, the curvature is defined as [39].

$$d\theta/dl = 1/r = l/A^2 + 1/R_{max}, \tag{1}$$

where l represents the curve length from the starting point (0, 0) with a curvature radius of R_{max} to position (p, q) with a curvature radius of R_{min}. r denotes the curvature radius. A is a positive constant, which is given by

$$A = \sqrt{\Delta L/(1/R_{min} - 1/R_{max})}, \tag{2}$$

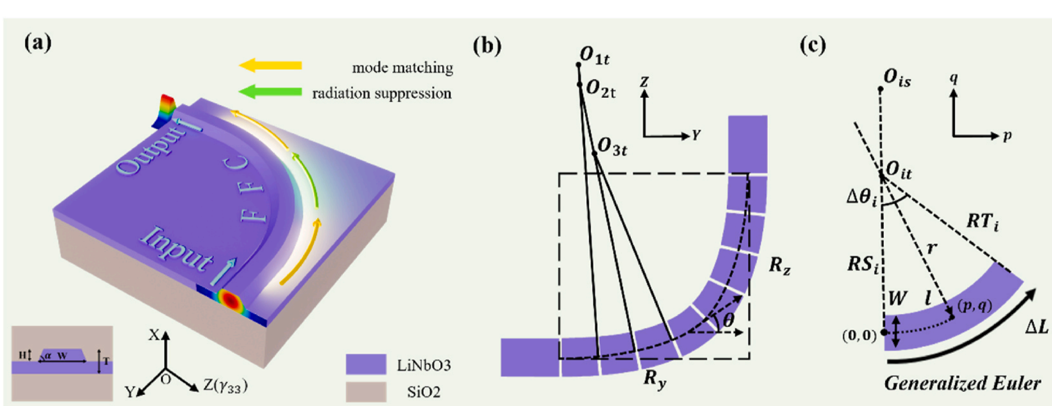

Fig. 1. (a) Perspective view of the proposed FFC design implemented using the FDB-MFA method on the LNOI platform. The inset illustrates the schematic of the waveguide cross-section. (b) Schematic diagram of the FFC structure, with equivalent radius R_{eq} defined as the maximum of R_y and R_z, representing the equivalent radius. (c) Schematic diagram of the Generalized Euler segment comprising the FFC structure.

which means that the curvature is always linearly increasing with curve length. Building on the concept of the modified Euler spiral, we develop the Generalized Euler spiral, which enables curvature to vary linearly—either increasing or decreasing—along the curve length. The curvature of the *i* th segment is expressed as

$$d\theta/dl = 1/r = l/a_i + 1/RS_i. \quad (3)$$

The value of the constant a_i is given by

$$a_i = \Delta L/(1/RT_i - 1/RS_i). \quad (4)$$

When $RT_i < RS_i$, the curve segment is a modified Euler spiral. It can be transformed to be with the Cartesian coordinate system, as shown in the following equations:

$$p = \int_0^l cos(s^2/2a_i + s/RS_i)ds, \quad (5)$$

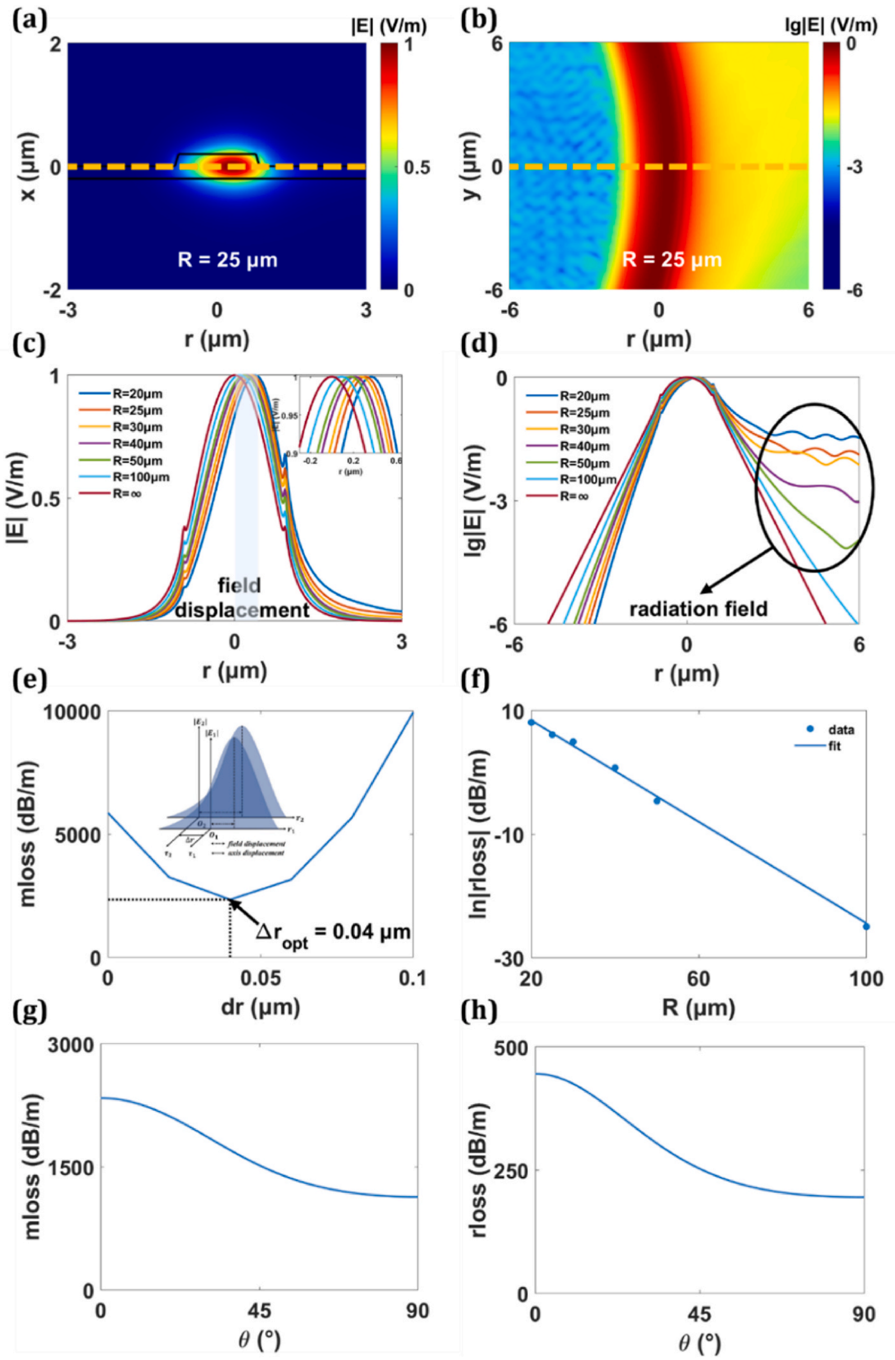

Fig. 2. (a) TE0 mode profile and (b) TE0 mode propagation in the bending waveguide with a curvature radius of 25 μm when the angle $\boldsymbol{\theta}$ is $\mathbf{0}^\circ$ at 1550 nm wavelength. (c, d) TE0 mode field distribution along the $\boldsymbol{r}$-axis with various curvature radii when the angle $\boldsymbol{\theta}$ is $\mathbf{0}^\circ$ at 1550 nm wavelength. (e) Calculated ***mloss*** between the mode fields for R = 25 μm and R = 30 μm when the angle $\boldsymbol{\theta}$ is $\mathbf{0}^\circ$ at 1550 nm wavelength. The inset illustrates the schematic of the axis displacement $\Delta\boldsymbol{r}$. (f) Calculated ***rloss*** for different R when the angle $\boldsymbol{\theta}$ is 0° at 1550 nm wavelength. A linear behavior is observed on a logarithmic scale of ***rloss***. (g) Calculated ***mloss*** for $\Delta\boldsymbol{r}$ = 0.04 μm between the mode fields for R = 25 μm and R = 30 μm and (h) calculated ***rloss*** for R = 25 μm at 1550 nm wavelength for different $\boldsymbol{\theta}$.

$$q = \int_0^l sin(s^2/2a_i + s/RS_i)ds. \tag{6}$$

Unlike previous studies that focus on optimizing the overall loss of bending waveguides by directly refining the three-dimensional model [40–42], our approach performs an optimization by calculating the loss contributions of each individual segment, which will be fast for further loss reduction by overlooking material anisotropy on the LNOI platform. Therefore, analyzing the loss sources in the bending waveguide is crucial for the optimization process. It is well known that the main loss in curved waveguides are mode mismatch loss and radiation loss when material absorption and sidewall scattering are neglected [44]. In order to further investigate these losses, we performed a detailed analysis by using the Finite Difference Eigenmode (FDE) solver and three-dimensional finite-difference time-domain (3D FDTD) solver. All designs in the study were based on an X-cut TFLN platform with a total film thickness T of 400 nm, and a buried oxide layer thickness of 4.7 μm. The waveguide sidewall angle α is approximately 70°, with the etching depth of 200 nm, determined by the fabrication process. The bottom width W of the bending waveguide is chosen to be 1.8 μm. At $\theta = 0°$, we set the curvature radius of the bending waveguide to 25 μm, and conducted simulations of the TE0 mode field distribution and the optical propagation mode within the bending waveguide at 1550 nm wavelength. The results are presented in Fig. 2(a) and 2(b). It is clear that the peak of TE0 mode in the bending waveguide is displaced along the positive r-axis relative to the center of the waveguide cross-section, which leads to mode mismatch loss. Moreover, during the bending process, a portion of the energy deviates from the original propagation path and radiates out of the waveguide, thereby contributing to radiation loss.

To quantitatively analyze these two losses on the X-cut LNOI platform, the TE0 mode intensity distribution was extracted along the r-axis at the center of the waveguide. By varying the curvature radius, we obtained different mode field intensity distributions. As shown in Fig. 2(c), the mode field displacement varies with the curvature radius, leading to different mode field distributions. Specifically, a smaller curvature radius corresponds to a larger mode field displacement. Based on the above analysis, we conclude that the transition between bending waveguides with different curvature radius introduces additional mode mismatch loss. We define the mode mismatch loss ($mloss_{i,j}$) between two arbitrary mode fields, $E_i(r,x)$ and $E_j(r,x)$, using the mode field overlap integral [45], as given by the following expression:

$$mloss_{i,j} = 10\, lg\left\{ \frac{\left[\iint E_i(r - \Delta r_{i,j}, x) E_j(r,x) drdx\right]^2}{\left[\iint E_i^2(r - \Delta r_{i,j}, x) drdx^* \iint E_j^2(r,x) drdx\right]} \right\} \Big/ \Delta L, \tag{7}$$

where $\Delta r_{i,j}$ represents the axis displacement of the waveguide corresponding to $E_i(r,x)$ relative to that of $E_j(r,x)$, as shown in the inset of Fig. 2(e). Notably, the spatial distributions of two mode fields in a bending waveguide are typically misaligned along the transmission axis, which significantly impacts *mloss*. In the FDE solver, the axis displacement is directly set as a parameter, referred to as *shift*. Using Equation (7), we calculated *mloss* between the mode fields for R = 25 μm and R = 30 μm with respect to Δr when the angle θ is 0° at 1550 nm wavelength as illustrated in Fig. 2(e). It can be seen that there exists an optimal Δr with a value of 0.04 μm (denoted as Δr_{opt}) that effectively compensates for the mode field displacement, ensuring optimal overlap between the two mode fields and minimizing *mloss*.

As shown in Fig. 2(d), the radiation field intensity is presented in logarithmic scale for ease of analysis. In the bending waveguide, the TE0 mode field no longer exhibits the typical exponentially decaying evanescent field along the positive r-axis, as seen in straight waveguides (bending radius is infinite). Instead, it manifests as an oscillatory radiative field. The smaller the curvature radius, the stronger the radiative field, leading to an increased radiation loss. According to bent waveguide theory [45], the expression of the radiation loss (*rloss*) with a bending radius of R is determined by

$$rloss(R) = C_1{}^*exp(-C_2{}^*R), \tag{8}$$

where C_1 and C_2 are constants independent of R. In the FDE solver, for a certain structure of waveguide, $rloss(R)$ can be directly calculated by substituting R as shown in Fig. 2(f). On a logarithmic scale of *rloss*, its relationship with R can be approximated as linear. Fig. 2(g) and 2(h) present the variations in *mloss* for $\Delta r = 0.04$ μm and $rloss(R)$ as θ ranges from 0° to 90°. The trends of *mloss* and $rloss(R)$ closely resemble those of the refractive index variation. This similarity arises from the anisotropic properties of the LNOI platform, where the refractive index varies with the propagation direction of light within the bending waveguide, resulting in the curvature-dependent mode field distribution influenced by the propagation angle θ.

According to the analysis of bending waveguide losses above, we propose the FDB-MFA method to modify and optimize the bending waveguide on the LNOI platform. Due to the anisotropic properties of the LNOI platform, we do not impose symmetry constraints. Instead, two separate optimization processes, using the FDE solver, are performed at the input and output ports of the bending waveguide, which are connected to the straight waveguides as shown in Fig. 3(a). As shown in Fig. 3(b), each optimization step only focuses on a single curve segment, progressively shaping the entire bending waveguide structure through iterative processes. To further reduce mode mismatch loss at the junction between the curves obtained from these two optimizations when they converge at a certain point along the bending waveguide, it is essential to satisfy the condition $|RT_{i_1} - RT_{i_2}| \le \Delta R$, where ΔR represents the curvature radius search precision, set to 100 nm to account for the manufacturing tolerance. Fig. 3(c) shows two scenarios in which the termination condition is not met. Taking $RT_{i_1} - RT_{i_2} > \Delta R$ as an example, optimization2 terminates prematurely, while optimization1 continues, progressively overwriting the results of optimization1 until the optimization termination condition is satisfied. The second scenario follows a similar optimization process. After finalizing these two optimization processes, we use a Generalized Euler to smoothly connect the endpoints of the two curves, as illustrated in Fig. 3(d).

As shown in Fig. 3(e), the FDB-MFA method is used to optimize the i th segment of FFC by finding the zero of the figure-of-merit (FOM_i) function, which is defined related with the losses.

$$FOM_i = \left(mloss_{i,i-1} < ML\right) \times \left(rloss(RT_i) < RL\right) - 0.5, \tag{9}$$

where ML and RL are the predefined thresholds for $mloss_{i,i-1}$ and $rloss(RT_i)$, respectively. In our implementation, the FOM is defined as a binary logic function that returns 1 when both the $mloss_{i,i-1}$ and $rloss(RT_i)$ are below their respective thresholds, and 0 otherwise. To enable zero-point detection using a bisection method, we subtract 0.5 from the binary result, such that the FOM only takes two values: −0.5 and 0.5. Although this constant has no direct physical meaning, it provides a symmetric value range centered around zero, which simplifies the search process. More generally, any value pair with opposite signs (e.g., in the range [−a, b], with a, b > 0) could serve the same purpose.

To ensure that the curvature radius RT_i decreases, a suitable minimum curvature radius R_{min} is set ($R_{min} = 20$ μm typically, or lower), and RT_i is then optimized within the initial interval [R_{min}, RT_{i-1}] using a bisection search algorithm. For the optimization of RT_1, RT_0 is set to a larger value ($RT_0 = 10000$ μm typically, or more). For clarity, the i th curve segment is denoted as S_iT_i. Although cross-sections S_i and T_{i-1} overlap, their corresponding curvature radii differ and must be treated as distinct. For each new value of RT_i, the mode mismatch loss occurs as the light enters from the cross-section T_{i-1} and propagates through the S_iT_i curve. This is caused by the field displacement of $E_i(r,x)$ relative to $E_{i-1}(r,x)$. Notably, the variation in the RT results in a corresponding axis displacement of $E_i(r,x)$ relative to $E_{i-1}(r,x)$, represented by $\Delta r_{i,i-1}$. By adjusting $\Delta r_{i,i-1}$, the optimal value of $\Delta r_{i,i-1}$ can be determined to

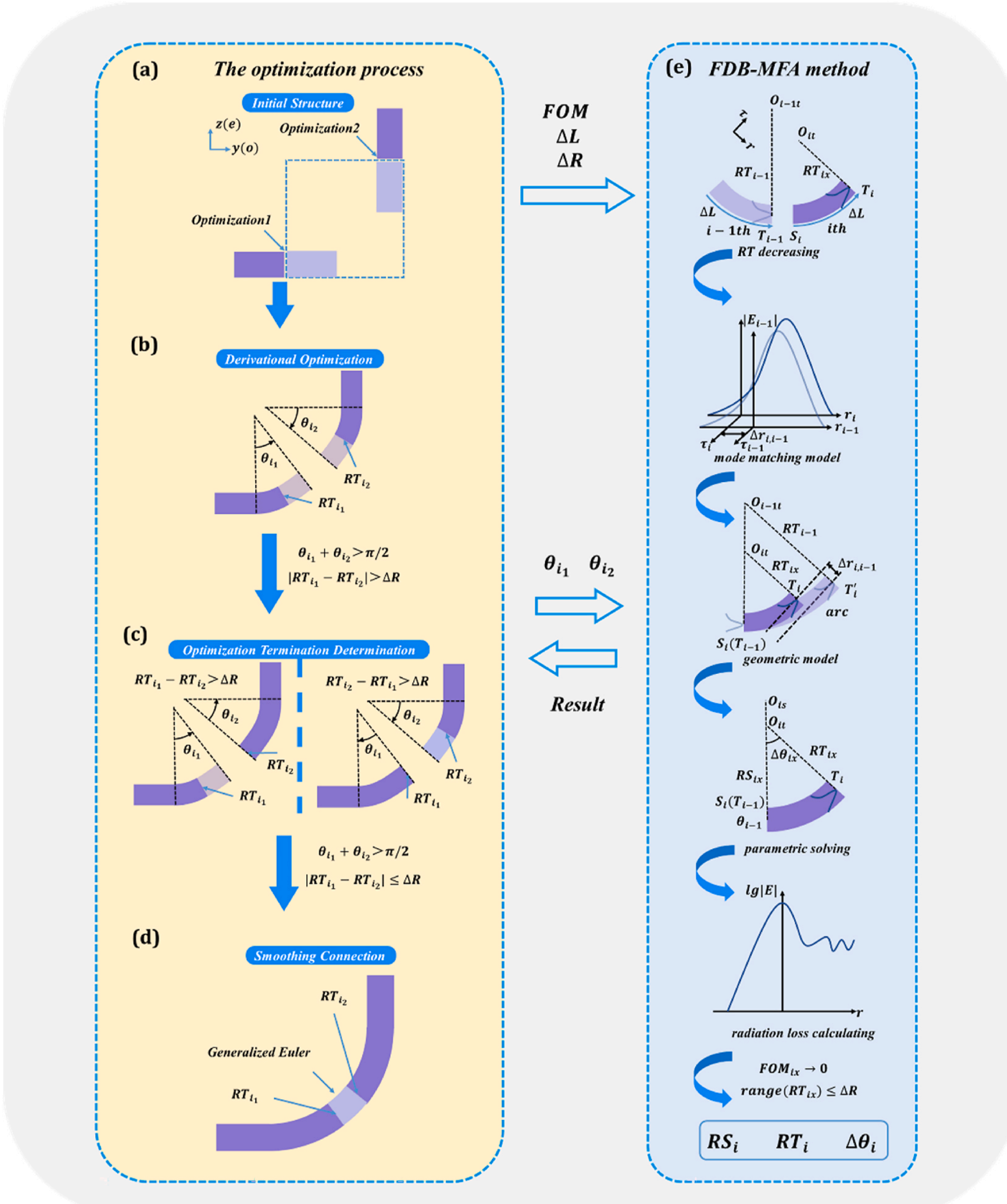

Fig. 3. (a–d) Schematic diagram of the optimization process. (e) The design flow of the FDB-MFA method.

minimize $mloss_{i,i-1}$, as illustrated in Fig. 2(e). In the FDE solver, a sequential scan is conducted over a wide interval [−100 nm, 100 nm] with a precision of 10 p.m. to ensure comprehensive search and determine the optimal value of $\Delta r_{i,i-1}$. To determine the parameters of the i th curve segment using $\Delta r_{i,i-1}$, the corresponding geometric model was established. Taking RT_{i-1} as the radius and $\Delta\theta_i$ as the angle, a circular arc S_iT_i' is constructed. Since $\Delta\theta_i$ is small, the refractive index is assumed to remain constant. The mode field at cross-section T_i' can be considered as the projection of the mode field at cross-section S_i. Therefore, $\Delta r_{i,i-1}$ is defined as the distance between the tangents at T_i and T_i', which is calculated by the following equation (See Supplementary Note 1 in Supporting Information for the proofs):

$$\Delta r_{i,i-1} = f(RT_{i-1}, RS_i, RT_i, \Delta\theta_i). \tag{10}$$

To ensure that RT_i does not decrease to the curvature radius associated with strong radiation fields, $rloss(RT_i)$ is also calculated. We employ a bisection search method to locate the zero of FOM_i, iteratively refining the search until the interval for RT_i reaches ΔR, at which point the result for the optimized i th curve segment is obtained and output.

The optimization process naturally divides into two stages. The first stage focuses on optimizing the mode matching region, where the primary objective is to reduce the bending size with minimal mode mismatch loss. During this phase, $mloss_{i,i-1}$ approaches the threshold ML, and the mode mismatch loss becomes the dominant factor in the optimization. The second stage focuses on optimizing the radiation suppression region, where the goal is to minimize radiation loss. As the curvature radius decreases, $rloss(RT_i)$ approaches the threshold RL, and the radiation loss becomes the primary optimization criterion.

3. Low-loss bend structures

Using the proposed design method, we optimize three low-loss bending waveguides with different dimensions by setting various values for ML and RL. The resulting equivalent radii are 33.9 μm, 40.5

μm and 45.5 μm, respectively (See Supplementary Note 2 in Supporting Information for the influence of *ML* and *RL* on the equivalent radius. See Supplementary Note 3 in the Supporting Information for the values of the design parameters for the optimized FFC). The FDE simulation for bending waveguide losses takes approximately 30 s per iteration, with mesh sizes set to 20 nm to ensure high-precision optimization. The final optimization results for each curve segment typically require around 15 min on a 24-core CPU server. In our designs, the number of segments ranges from 14 to 23 depending on the target bending radius, resulting in a total optimization time of approximately 1.8–3 h per structure. For comparison, an inverse-designed bend on the SOI platform [42] employed a 64-core server and required approximately 18 h of FDTD-based optimization, even for a bend with an equivalent radius of only 9.35 μm. These results demonstrate the computational efficiency and scalability of our proposed FDB-MFA method. The performance of the curves is verified through 3D FDTD simulations. Fig. 4(a)-4(c) show the simulation results for the transmission efficiency of the optimized FFC bending waveguides with different sizes. The inset shows the dimensional model of the corresponding FFC bends. We also compare the transmission performance with that of traditional curve designs having the same equivalent radius, including Bezier bend, Euler bend, and Arc bend. For the Bezier bend, the optimal design is achieved with a Bezier number $B = 0.35$ [39]. For the Euler bend, the minimum radius of 18.1 μm, 21.6 μm and 24.3 μm are chosen to match the equivalent radius of the other curves. For the optimized FFC bends with equivalent radii of 33.9 μm, 40.5 μm and 45.5 μm, the transmission losses are below 0.022 dB, 0.012 dB and 0.004 dB, respectively, over the 100 nm band. At the specific operating wavelength of 1550 nm, the corresponding losses are 0.015 dB, 0.0074 dB, and 0.0012 dB, respectively. For all three sizes, the optimized FFC bends demonstrate superior transmission performance compared to the traditional design curves. We also employed FDTD simulations to extract the variation of the net TE0 mode power content as a function of transmission length for the optimized FFC, Bezier, Euler, and Arc bends, all with an equivalent radius of 33.9 μm. This analysis further demonstrates that the optimized FFC bend exhibit superior TE0 mode propagation, enabling low loss single mode transmission (See Supplementary Note 4 in the Supporting Information).

To evaluate the permitted manufacturing tolerance for the designed bending waveguide, we introduce a width deviation ΔW by moving each point on the boundary shape curve along the outer normal vector at a distance of $\Delta W/2$. The value of ΔW varies from −200 to 200 nm with a 20 nm interval. The simulated transmission efficiencies of the optimized FFC and other traditional designs at 1550 nm wavelength vary with ΔW, as shown in Fig. 4(d)-4(f). Within the manufacturing tolerance range of −200 to 200 nm, the maximum transmission losses for the FFC bends with equivalent radii of 33.9 μm, 40.5 μm and 45.5 μm are 0.026 dB, 0.008 dB and 0.003 dB, respectively, all of which are lower than the minimum transmission losses for the traditional designs with the same size. This demonstrates that the proposed FFC design exhibits excellent manufacturing tolerance.

The designed devices were fabricated on a commercially available X-cut LNOI wafer and tested. An *i*-line stepper was used for patterning, followed by inductively coupled plasma etching to transferred onto the LN layer. Finally, a 0.8 μm thick silicon dioxide upper cladding was deposited onto the structure using plasma-enhanced chemical vapor deposition. The Microscopic and SEM image of the fabricated structures with equivalent radius of 33.9 μm are illustrated in Fig. 5(a)-5(d). To minimize random errors caused by measurement instability, 80 cascaded FFC bends, Bezier bends, Euler bends, and Arc bends were fabricated. All test ports were connected to TE-polarized grating couplers (GCs), which had an insertion loss of approximately 7 dB/facet at 1550 nm for optical coupling between the chip and the fiber. The actual fabricated waveguide bottom width was 1.83 μm, with the same equivalent radii for each bend. The measured transmission losses in the wavelength range of 1500–1600 nm for each port, divided by 80 cascaded bends, are shown in Fig. 5(e)-5(g). It can be seen that for FFC bends with equivalent radii of 33.9 μm, 40.5 μm and 45.5 μm, the maximum losses are 0.034 dB, 0.019 dB and 0.02 dB, respectively. At 1550 nm wavelength, the transmission losses for FFC bends with equivalent radii of 33.9 μm, 40.5 μm and 45.5 μm are 0.022 dB, 0.008 dB and 0.009 dB, respectively. These transmission performances significantly outperform the Bezier, Euler, and Arc bends of the same dimensions. The measured transmission efficiencies are lower than the simulated results, which is possibly caused by the waveguide sidewall etching roughness and width fabrication deviation during the manufacturing process. Furthermore, the measured losses of the FFCs with equivalent radii of 40.5 μm and 45.5 μm are nearly identical. Based on the simulation results, it is likely attributed to the minimal contributions of mode mismatch loss and radiation loss, with sidewall scattering emerging as the dominant loss mechanism.

4. Compact and low-loss ODLs

We apply the proposed method to design a compact and low-loss

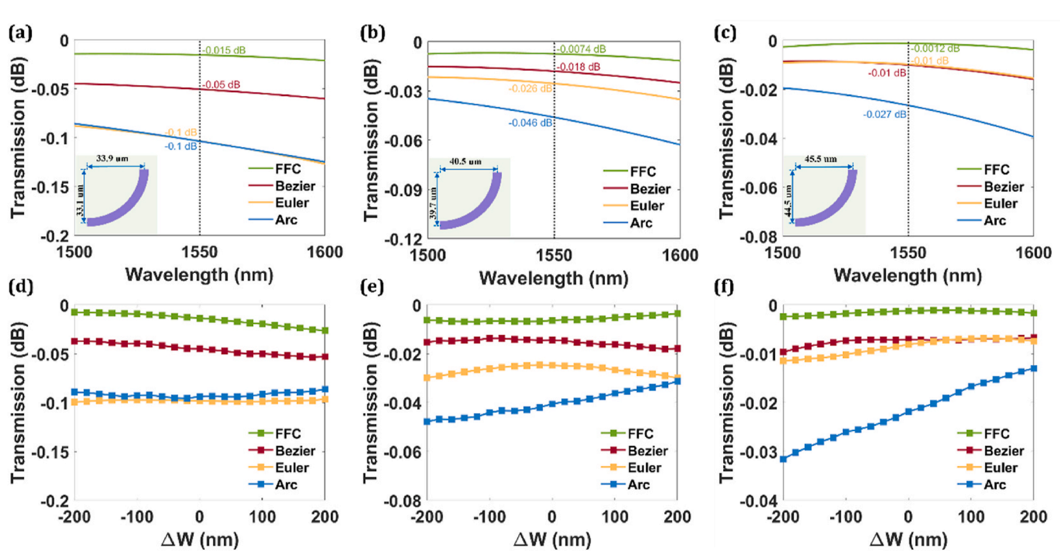

Fig. 4. Simulated transmission spectra for optimized FFC, Bezier, Euler, and Arc bends with (a) $R_{eq} = 33.9$ μm, (b) $R_{eq} = 40.5$ μm and (c) $R_{eq} = 45.5$ μm. The inset shows the dimensional model of the corresponding FFC bends. Simulated transmission efficiencies at 1550 nm wavelength for optimized FFC, Bezier, Euler, and Arc bends with (d) $R_{eq} = 33.9$ μm, $R_{eq} = 40.5$ μm and (f) $R_{eq} = 45.5$ μm, as a function of the width deviation ΔW.

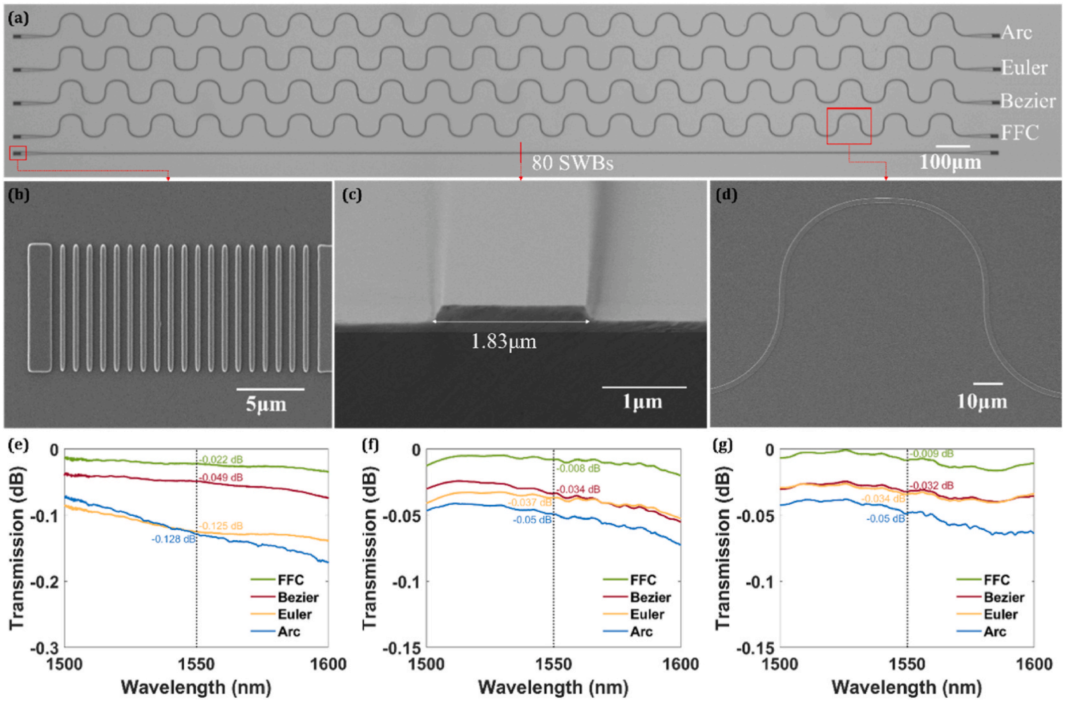

Fig. 5. (a) Microscopic image of the fabricated devices with 80 cascaded bends and an equivalent radius $\boldsymbol{R_{eq}} = 33.9$ μm. (b) SEM images of (b) the grating coupler, (c) waveguide structure and (d) optimized FFC with $\boldsymbol{R_{eq}} = 33.9$ μm. Measurement results of the normalized transmittances for optimized FFC, Bezier, Euler, and Arc bends with (e) $\boldsymbol{R_{eq}} = 33.9$ μm, (f) $\boldsymbol{R_{eq}} = 40.5$ μm and (g) $\boldsymbol{R_{eq}} = 45.5$ μm, which are the average values out of 80 fabricated bends.

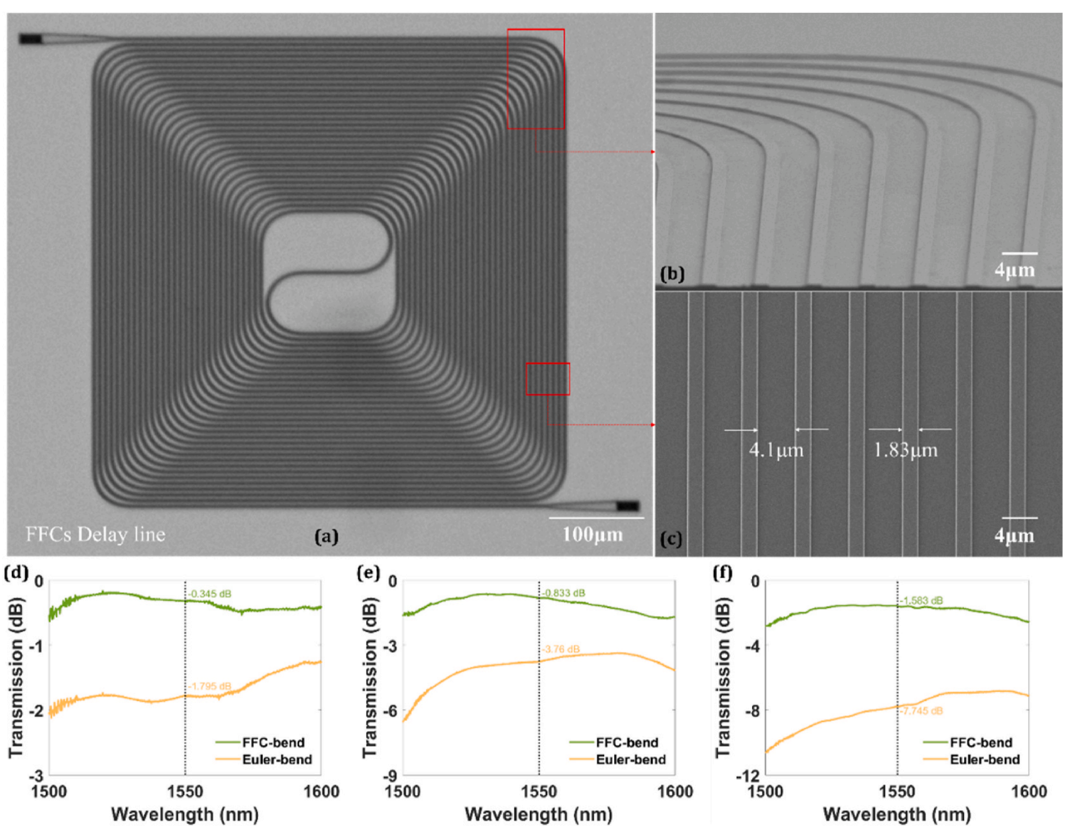

Fig. 6. (a) Microscopic image of the fabricated FFC-bend ODLs with the total length of 3.64 cm. SEM images of (b) the FFC-bends and (c) the straight sections of the fabricated FFC-bend ODLs. Transmission measurements of (d) 0.76 cm, (e) 1.96 cm and (f) 3.64 cm long for the FFC-bend ODLs and the Euler-bend ODLs on the same chip.

ODLs. To balance loss and size, we select a gap of 4.2 μm to avoid mode coupling between adjacent waveguides and optimize an S-bend waveguide with an equivalent radius of 30 μm (See Supplementary Note 5 in the Supporting Information). Additionally, the 90-degree optimized FFC bending waveguide with the equivalent radius of 45.5 μm is used to form the ODL structure.

The fabrication process of the designed delay line follows the same procedure as that of the FFC bending waveguide. Fig. 6(a)-6(c) show the Microscopic and SEM images of the optimized FFC bending delay line sample, with a total length of 3.64 cm and an area footprint of approximately 0.5 $\times$ 0.5 mm^2. The fabricated waveguide has a bottom width of 1.83 μm, with a gap of 4.1 μm between neighbored waveguides. Although the waveguide gap is slightly smaller than the design value, it still effectively prevents mode coupling. Three ODLs with lengths of 0.76 cm, 1.96 cm and 3.64 cm were designed. For comparison, ODLs with the same equivalent radius using Euler bends were also fabricated on the same chip. The measured transmission losses were normalized by comparing them to the transmission loss of a 500 μm long straight waveguide with the same grating couplers, and the propagation loss (excluding coupling loss) is shown in Fig. 6(d)-6(f). The results show that the transmission loss of the FFC-bend ODLs is significantly lower than that of the Euler-bend ODLs of the same dimensions. Specifically, for the FFC-bend ODLs with total lengths of 0.76 cm, 1.96 cm and 3.64 cm, the transmission losses at 1550 nm wavelength are 0.345 dB, 0.833 dB and 1.583 dB, respectively, corresponding to average transmission losses of 0.45 dB/cm, 0.43 dB/cm and 0.43 dB/cm. In contrast, the Euler-bend ODLs of the same lengths exhibit transmission losses of 1.795 dB, 3.76 dB and 7.745 dB at 1550 nm wavelength, with average transmission losses of2.36 dB/cm, 1.92 dB/cm and 2.13 dB/cm, respectively. Compared to the Euler-bend ODLs, the transmission loss of the FFC-bend ODLs is reduced by approximately 80 % for the same sizes.

5. Conclusion

In conclusion, we propose the FDB-MFA method for designing compact and low-loss SWBs on anisotropic LNOI platform. By optimizing the FFC structure consisting of cascaded Generalized Euler segments, we overcome the loss limitations imposed by the anisotropic platform, significantly reducing both mode mismatch loss and radiation loss. SWBs utilizing FFCs with a 200-nm-thick slab and a 200-nm-height rib on the X-cut LNOI platform, designed with three distinct equivalent radii, outperform traditional bending structures and exhibit a substantial manufacturing tolerance of $\pm$200 nm. Experimental results show that the bending waveguide with an equivalent radius of 40.5 μm exhibits a transmission loss of 0.008 dB at 1550 nm wavelength. Additionally, we demonstrate a compact FFC-bend optical delay line (ODL) with a total length of 3.64 cm, a footprint of 0.5 $\times$ 0.5 mm^2, and a transmission loss of 0.43 dB/cm, 80 % lower than the Euler-bend ODLs of the same size. The low-loss SWBs and ODLs developed in this work offer considerable potential to drive the advancement of LNOI-based photonic integrated devices and circuits.

CRediT authorship contribution statement

Lingxiao Zhang: Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Minjun Xie:** Formal analysis, Data curation, Conceptualization. **Yalong Yu:** Data curation, Conceptualization. **Juzheng Li:** Data curation, Conceptualization. **Wei Ma:** Conceptualization. **Tao Chu:** Conceptualization. **Bo Xiong:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.optcom.2025.132032.

Data availability

Data will be made available on request.

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